CE 310 — Week 3 Pre-Lab Quiz

Probability Basics: Marginal, Conditional, Joint, and Independence

Due: Before Week 3 Lab — submit via D2L Total: 10 points (1 point each) Estimated time: ~10 minutes

Instructions: Choose the single best answer for each question. This quiz is auto-graded and must be completed before attending the Week 3 lab session. You may use your notes and readings, but work independently.

Q1. An engineer wants to know whether high cooling demand is equally likely year-round. She calculates P(Cooling > 50 kWh) = 14.21% across all 8,760 hours. Then she calculates P(Cooling > 50 kWh | July) = 38.31%. What does this difference reveal?

(A) The original 14.21% estimate contains a calculation error that must be corrected before use.
(B) Cooling demand is conditionally dependent on month — pooling all months hides the elevated seasonal risk.
(C) July data is an outlier and should be excluded from the annual probability estimate.
(D) Marginal and conditional probabilities must always agree when the sample is large enough.

Q2. Before the 1986 Challenger launch, engineers had recorded O-ring failure data from previous flights. The critical mistake in the pre-launch analysis was:

(A) Using too small a sample — fewer than 10 prior flights had been analyzed.
(B) Reporting probabilities as decimals instead of percentages, causing a miscommunication.
(C) Analyzing failures without conditioning on launch temperature, obscuring the temperature-failure relationship.
(D) Confusing joint probability with marginal probability when combining failure modes from two boosters.

Q3. A design standard states: "Size the cooling system for peak-hour demand." Which probability concept best justifies conditioning on summer months and business hours when estimating that peak demand?

(A) The law of large numbers — larger samples converge to the true population mean regardless of conditioning.
(B) The addition rule — P(A or B) accounts for overlap between summer and business-hours subsets.
(C) Conditional probability — the distribution of demand given the operating context is more relevant to design than the unconditional average.

(D) Independence — summer hours and business hours are independent events, so their probabilities multiply.

Q4. The dataset used in this week's lab contains hourly energy readings for a medium office building in Climate Zone 2B (Tucson area) for the year 2023. How many total rows of data are in the dataset?

(A) 365
(B) 4,380
(C) 8,760
(D) 12,960

Q5. July is represented in the dataset by exactly 744 hourly rows. Which calculation correctly gives the marginal probability that a randomly selected hour falls in July?

(A) 744 / 365
(B) 744 / 12
(C) 744 / 8,760
(D) 8,760 / 744

Q6. Which statement correctly distinguishes marginal probability from joint probability?

(A) Marginal probability applies only to continuous variables; joint probability applies only to discrete variables.
(B) Marginal probability gives the likelihood of a single event across the full sample; joint probability gives the likelihood that two events both occur simultaneously.
(C) Marginal probability is computed from a histogram; joint probability is computed from a scatter plot.
(D) Marginal and joint probability are interchangeable terms when events are mutually exclusive.

Q7. Two events A and B are statistically independent. Which of the following equalities must hold?

(A) $P(A \mid B) = P(B)$
(B) $P(A \text{ and } B) = P(A) + P(B)$
(C) $P(A \mid B) = P(A)$
(D) $P(A \text{ and } B) = 0$

Q8. A building fault-detection system uses a Naive Bayes classifier to flag hours when equipment may be malfunctioning. The classifier estimates P(Electricity > 150 kWh | Fault) and P(Electricity > 150 kWh | Normal) from historical data, then applies Bayes' theorem to calculate the probability of a fault given the observed reading. Which statement best describes what the classifier is doing?

(A) Computing marginal probabilities and comparing them to a fixed threshold set by the equipment manufacturer.
(B) Using conditional probability — the likelihood of observing a high electricity reading given each class — to update the probability that a fault is occurring.
(C) Checking statistical independence between electricity consumption and fault status before issuing an alert.
(D) Applying the addition rule to combine fault probabilities from multiple sensors into a single score.

Q9. In Excel, you have a column labeled Season and a column labeled Cooling_kWh. You want to count the number of hours where Season = “Summer” AND Cooling_kWh > 50. Which formula is correct?

(A) =COUNTIF(Season,"Summer", Cooling_kWh,">50")
(B) =COUNTIFS(Season,"Summer", Cooling_kWh,">50")
(C) =SUMIF(Season,"Summer", Cooling_kWh,">50")
(D) =AVERAGEIFS(Cooling_kWh, Season,"Summer", Cooling_kWh,">50")

Q10. You want to find the average electricity consumption during business hours only. Your data has a BusinessHours column containing “Yes” or “No” and an Electricity_kWh column. Which Excel formula returns the correct result?

(A) =AVERAGE(IF(BusinessHours="Yes", Electricity_kWh))
(B) =COUNTIF(BusinessHours,"Yes", Electricity_kWh)
(C) =AVERAGEIF(BusinessHours,"Yes", Electricity_kWh)
(D) =SUMIFS(Electricity_kWh, BusinessHours,"Yes") / COUNT(Electricity_kWh)